ROUTH-HURWITZ: FINDING THE CRITICAL GAIN

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Build the Routh array for a third-order angle control loop and find the range of proportional gains that keeps it stable.	☐
2	Predict the critical gain and the frequency of oscillation at that gain, and measure both on the DC motor.	☐
3	Explain why the loop can become unstable once a measurement filter is added, and why the measured limit differs from the prediction.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The angle of the shaft is measured from the encoder counts, as described in the practical exercise, Manual PID Tuning of DC Motor Angle Control, and the output voltage of the controller sets both the direction and the PWM value. Because the angle is the integral of the speed, the angle model is $G_\theta(s) = K_\theta/(s(\tau s + 1))$, where $K_\theta = 6K$ converts RPM/V into degrees per second per volt. For the A1.4 example motor, $K_\theta = 168$ °/(s·V).

Safety: this exercise deliberately takes the loop to the edge of instability. Keep the output saturation at ±5 V, use small setpoint steps, keep a stop switch in the model, and stop the test as soon as a sustained oscillation appears.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A first-order low-pass filter is added to the angle measurement, making the closed loop

third order. The Routh-Hurwitz criterion predicts the gain at which it becomes unstable, and the prediction is tested by raising the gain carefully on the hardware. The exercise supports the ACE-CORE learning guide, Stability and the Routh-Hurwitz Stability Criterion.

Build the Routh Array

With the angle model and a filter $1/(\tau_f s + 1)$ in the feedback path, the characteristic equation of the closed loop is:

$$\tau\tau_f\, s^3 + (\tau + \tau_f)\, s^2 + s + K_p K_\theta = 0$$

All coefficients are positive for $K_p > 0$. The Routh array is:

Table 2: *Routh Array for the Angle Loop with a Measurement Filter*

Row	Column 1	Column 2
s^3	$\tau\tau_f$	1
s^2	$\tau + \tau_f$	$K_p K_\theta$
s^1	$1 - \tau\tau_f K_p K_\theta/(\tau + \tau_f)$	0
s^0	$K_p K_\theta$	

The loop is stable while every entry in the first column is positive, which gives the stable range and the critical gain. At the critical gain the s^1 row is zero, and the auxiliary equation from the s^2 row gives the frequency of the sustained oscillation:

$$0 < K_p < K_{p,\,crit} = \frac{\tau + \tau_f}{\tau\tau_f\, K_\theta}, \qquad \omega_{osc} = \frac{1}{\sqrt{\tau\tau_f}}$$

Without the filter ($\tau_f = 0$), the loop is the second-order system of the practical exercise, Second-Order Response: DC Motor Angle under P Control, and is stable for every positive gain. The filter adds a third pole, and with it a gain limit. Predicted values for the example motor are given in Table 3.

Measure the Critical Gain

1. Add a first-order low-pass filter with $\tau_f = 0.05$ s to the angle measurement in the model from the practical exercise, Second-Order Response: DC Motor Angle under P Control.
2. Calculate $K_{p,crit}$ and ω_{osc} for your motor.
3. Starting at about 20% of $K_{p,crit}$, apply a 10° setpoint step and observe the response. Increase K_p by about 10% each time.
4. Stop at the first gain for which the oscillation no longer dies away. Record this gain and measure the period of the oscillation.
5. Repeat with $\tau_f = 0.1$ s.

Example responses for $\tau_f = 0.05$ s are shown in Figure 1. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares ($K = 28$ RPM/V, $\tau = 0.15$ s), so they show the behaviour to expect rather than measured data. At 70% of the predicted critical gain, the simulated loop is only lightly damped and oscillates for several seconds.

The simulation assumes the controller output takes effect one full sample after the measurement, a pessimistic case, and with this delay the simulated loop becomes unstable at about 79% of the continuous-time prediction for τ_f = 0.05 s, and about 86% for τ_f = 0.1 s. Expect your measured limit to lie between the simulated value and the prediction.

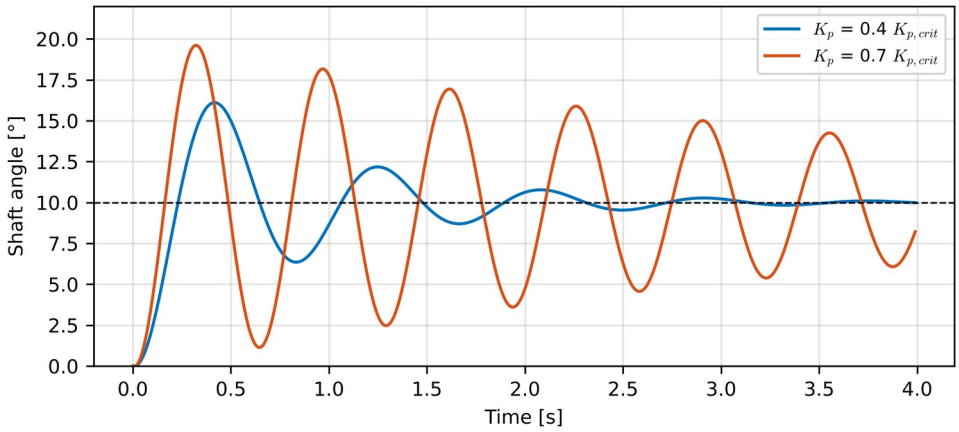


Figure 1: *Simulated Example of the Angle Response to a 10° Step at 40% and 70% of the Predicted Critical Gain, with τ_f = 0.05 s*

Table 3: *Predicted and Measured Critical Gains*

τ_f (s)	Predicted $K_{p,crit}$ (V/°)	Predicted ω_{osc} (rad/s)	Predicted period (s)	Measured $K_{p,crit}$ (V/°)	Measured period (s)
0.05	0.159	11.5	0.54		
0.1	0.099	8.2	0.77		

The predicted values in Table 3 use the example motor. The measured critical gain is usually close to the prediction. Friction adds damping and can raise it, while the sampling delay of the digital controller lowers it. Larger filter time constants lower the critical gain and slow the oscillation.

1. Why was the angle loop stable for every gain without the filter, but not with it?
2. What does a row of zeros in the Routh array tell you, and how is the auxiliary equation used?
3. How much margin below the critical gain would you leave in a real product, and why?

4.0 Concluding Remarks

This practical exercise used the Routh-Hurwitz criterion to find the range of stable gains for a third-order angle control loop and tested the result on hardware. Using MATLAB, Simulink and ACE-Lab hardware, the critical gain and oscillation frequency were predicted from the Routh array and measured by raising the gain until the loop oscillated.

Adding a measurement filter introduced a third pole and with it a limit on the gain, showing that every extra lag in a loop reduces the gain it can tolerate. The Routh

array gives this limit, and the frequency of the resulting oscillation, from the characteristic equation alone, without solving for the poles.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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